Sergio Ruiz Carmona

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Summary

Expert RWE Data Analyst with a PhD in Biomedicine and 8+ years of experience in clinical analytics and observational research. Specialized in extracting and modeling secondary patient data from complex EMR systems (Epic) using SQL, R, and Python. Proven track record in descriptive cohort analysis, disease epidemiology tracking, and developing interactive BI dashboards to inform key medical decisions. Experienced in mentoring multidisciplinary teams and collaborating with pharmaceutical partners to translate complex real-world data into actionable clinical insights.

Skills

RWE Analytics: Observational Research Design, Descriptive Cohort Analysis, Disease Epidemiology, Survival Analysis, Patient Feasibility

Data Engineering: SQL Expert, ETL Processing, Epic EMR, Secondary Data Sources, Electronic Health Records (EHR)

Programming: R (Tidyverse, Shiny), Python (Pandas, NumPy), PySpark, Bash, Git

Data Visualization: Power BI, R Shiny, Interactive Dashboards, Business Intelligence Stacks

Domain Expertise: Biostatistics, Bioinformatics, Clinical Data Management, Pharmaceutical R&D

Experience

Center for Child Health Analytics, Luzerner Kantonsspital, Data Analyst — Luzern, CH
Mar 2025 – present

- Extract and model **secondary patient data** across complex Epic EMR systems containing millions of rows and relational tables.
- Engineer **statistical pipelines** using R and Python for guideline monitoring and safety analyses to verify patient outcomes.

Centre for Health Analytics, Royal Children's Hospital, Clinical Data Analyst — Melbourne, AUS
Oct 2022 – Feb 2025

- Formulated complex **SQL extraction protocols** to query multi-million-row EHR databases for observational research and cohort identification.
- Deployed custom **clinical dashboards** for tracking chronic conditions, acting as an SME for institutional clinical analytics.

Baker Heart and Diabetes Institute, Senior Research Officer and Bioinformatician — Melbourne, AUS
Jan 2019 – Oct 2022

- Led data science operations for 10 research groups, managing **statistical data analysis** for high-profile projects including UK Biobank.
- Acted as an **analytical mentor** and guide for new personnel, conducting technical workshops to build core programming competencies.

Universitat de Barcelona, Postdoctoral Researcher — Barcelona, ES
Jan 2017 – Dec 2018

- Headed computational oncology research focusing on **descriptive cohort tracking** and target selection in collaboration with hospitals.
- Managed **industry-funded partnerships** with pharmaceutical firms like Servier to validate drug target properties via simulations.

Education

PhD **Universitat de Barcelona**, Biomedicine — Barcelona, ES — July 2012 – Jan 2017

- Developed and validated novel structure-based computational drug discovery and virtual screening methodologies, scaling pipelines to screen libraries of over 1.1 million compounds to identify novel mechanisms of action.
- Translated baseline computational research into high-value assets, directly enabling a registered oncology drug patent and a biotech spin-off company targeting blood cancers.
- Excellent cum Laude. Awarded the Extraordinary Prize for PhD Thesis and the Ramon Margalef Award for best scientific publication.

MSc **Universitat Pompeu Fabra**, Bioinformatics for Health Sciences — Barcelona, ES — Sept 2010 – June 2012

- Designed and executed rational computational workflows targeting the druggable genome, focusing on structural protein-protein interfaces as a novel target class.
- Developed algorithmic approaches to isolate and predict interfacial binders, establishing the primary methodology that directly scaled into a successful doctoral thesis and subsequent patent framework.

Publications

Evaluating the incidence of complications among people with diabetes according to age of onset: Findings from the UK Biobank — June 2024
Seyedeh Forough Sajjadi, *Sergio Ruiz-Carmona*, Jonathan E Shaw, Dianna J Magliano
(Diabetic Medicine)

COVID-19 thromboembolism is reduced in ambulatory, but not hospitalized patients, following COVID-19 vaccination — Feb 2023
Hannah Stevens, *Sergio Ruiz-Carmona*, Karlheinz Peter, James D McFadyen
(Hematologica)

Oxygen Pathway Limitations in Patients with Chronic Thromboembolic Pulmonary Hypertension — Apr 2021
Erin J Howden*, *Sergio Ruiz-Carmona**, Andre La Gerche, Marion Delcroix, Guido Claessen
(Circulation)

Dynamic undocking and the quasi-bound state as tools for drug discovery — Jan 2017
Sergio Ruiz-Carmona, Peter Schmidtke, Rod Hubbard, Xavier Barril
(Nature Chemistry)

Selected Activities

Structured medical diagnoses and their free-text overwrite in EHR — Jan 2026

- Presented analysis of EHR data structures at Padiatrie Schweiz Conference.

Australasian Leadership Computing Grants — Sept 2020

- At the Baker Institute lab, I was awarded a 1 year computing grant by the Australian National Computational Infrastructure to study COVID-19 proteins and possible drug treatments

Compounds for use in the treatment of hyperproliferative disorders — Dec 2018

- Registered oncology drug patent based on computational cohort findings and structural biology.